

Problem

A three-bit DAC is shown in Fig. 5.37.

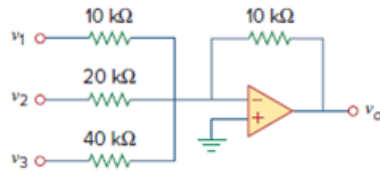
(a) Determine $|V_o|$ for $[V_1 V_2 V_3] = [010]$.

(b) Find $|V_o|$ if $[V_1 V_2 V_3] = [110]$.

(c) If $|V_o| = 1.25$ V is desired, what should be $[V_1 V_2 V_3]$?

(d) To get $|V_o| = 1.75$ V, what should be $[V_1 V_2 V_3]$?

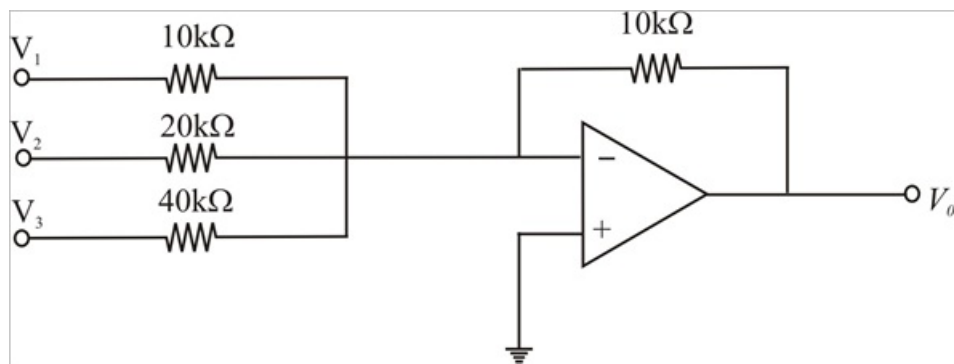
Figure. 5.37: Three-bit DAC



Step-by-step solution

Step 1 of 6

Given three bit DAC is



Step 2 of 6

(a)

Given circuit is inverting summing amplifier

$$V_o = -\left(\frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2 + \frac{R_f}{R_3} V_3\right)$$

$$V_o = -\left(\frac{10k}{10k} V_1 + \frac{10k}{20k} V_2 + \frac{10k}{40k} V_3\right)$$

$$V_o = -(V_1 + 0.5V_2 + 0.25V_3)$$

Step 3 of 6

Given

$$[V_1 \ V_2 \ V_3] = [0 \ 1 \ 0]$$

$$V_0 = -(0 + 0.5 \times 1 + 0)$$

$$V_0 = -0.5$$

$$|V_0| = |-0.5|$$

$$|V_0| = 0.5V$$

Step 4 of 6

(b) Given

$$[V_1 \ V_2 \ V_3] = [1 \ 1 \ 0]$$

From the equation we have

$$V_0 = -(V_1 + 0.5V_2 + 0.25V_3)$$

$$V_0 = -(1 + 0.5 \times 1 + 0)$$

$$V_0 = -1.5$$

$$|V_0| = |-1.5|$$

$$|V_0| = 1.5V$$

Step 5 of 6

Given

$$|V_0| = 1.25V$$

The equation we have is

$$V_0 = -(V_1 + 0.5V_2 + 0.25V_3)$$

In the above equation choose

$$V_1 = 1$$

$$V_2 = 0$$

$$V_3 = 1$$

$$1.25 = -(1 + 0 + 0.25 \times 1)$$

$$1.25 = -(1.25)$$

Thus we have

$$\text{For } |V_0| = 1.25 \text{ V}$$

$$[V_1 \ V_2 \ V_3] = [1 \ 0 \ 1]$$

Step 6 of 6

(d)

Given $|V_0| = 1.75\text{V}$

The equation we have is

$$V_0 = -(V_1 + 0.5V_2 + 0.25V_3)$$

In the above equation choose

$$V_1 = 1$$

$$V_2 = 1$$

$$V_3 = 1$$

$$1.75 = -(1 + 0.5 \times 1 + 0.25 \times 1)$$

$$1.75 = -(1.75)$$

Thus we have

$$\text{For } |V_0| = 1.75 \text{ V}$$

$$[V_1 \ V_2 \ V_3] = [1 \ 1 \ 1]$$